

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduct:

I Shaik Mubarak(192425194) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shaik Mubarak

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

1. Problem Analysis and Agent Design

Problem Statement

Universities receive numerous student queries related to course registration, examination schedules, timetables, attendance, and campus services. Handling these queries manually is time-consuming and increases the workload of administrative staff. An AI-based Student Support Assistant developed using Dialogflow can automate responses, provide instant information, and improve student satisfaction.

Introduction

Artificial Intelligence (AI) enables machines to perform tasks such as understanding language, learning, and decision-making. A Student Support Assistant is an AI-powered chatbot that uses Natural Language Processing (NLP) to understand student queries and provide relevant responses. It improves communication by offering accurate and 24×7 support.

Problem Analysis

The existing student support system has several limitations:

- Manual handling of repetitive queries.
- Delayed responses during peak periods.
- High workload for administrative staff.
- Possibility of human errors.
- Limited availability during office hours.

The proposed chatbot solves these issues by providing automated and accurate responses.

How AI Solves the Problem

The chatbot uses:

- Natural Language Processing (NLP): Understands user queries.
- Intent Recognition: Identifies the user's request.
- Entity Extraction: Extracts details such as course, semester, or date.
- Machine Learning: Improves response accuracy over time.
- Context Management: Maintains conversation continuity.

Objectives

- Provide 24×7 student support.
- Answer course registration queries.
- Display examination schedules.

- Provide timetable information.
- Check attendance details.
- Offer campus service information.
- Reduce administrative workload.
- Improve student satisfaction.

Agent Design

Users

- Students
- Faculty
- Administrative Staff
- New Admissions

Environment

- Dynamic
- Real-time
- Cloud-based
- Multi-user

Inputs

- Text messages
- Voice commands
- Button selections

Outputs

- Course registration details
- Examination schedules
- Timetable
- Attendance information
- Campus services
- Important announcements

Agent Architecture

**User → Dialogflow → NLP → Intent Recognition → Entity Extraction → Response
Generation → User**

Performance Measures

Measure	Description
Accuracy	Correct responses
Response Time	Fast reply
User Satisfaction	Positive feedback
Availability	24×7 service
Completion Rate	Successful query resolution

Advantages

- Available 24×7.
- Instant responses.
- Reduces administrative workload.
- Improves communication.
- Handles multiple users simultaneously.
- Reduces human errors.

Limitations

- Cannot handle highly complex queries.
- Requires internet connectivity.
- Needs regular updates.

Conclusion

The AI-based Student Support Assistant improves university support services by providing fast, accurate, and automated responses. It enhances communication, reduces manual effort, and offers a better experience for students.

2. Dialogflow Agent Development

Problem Statement

Develop a conversational chatbot using Google Dialogflow that understands student queries, identifies user intents, extracts important entities, and provides appropriate responses regarding university services.

Introduction

Dialogflow is Google's cloud-based conversational AI platform that allows developers to build intelligent chatbots using Natural Language Understanding (NLU). It processes user queries, identifies their intent, extracts important information, and generates accurate responses.

Objectives

- Develop an AI-powered chatbot.
- Understand natural language queries.
- Identify user intents.
- Extract entities.
- Generate meaningful responses.
- Maintain conversation context.
- Improve response accuracy.

Dialogflow Components

1. Agent

The Agent is the complete chatbot that processes user requests and generates responses.

Agent Name: Student Support Assistant

2. Intents

Intents define the purpose of user queries.

Intent	Purpose
Welcome	Greeting
Course Registration	Registration details
Timetable	Class schedule
Examination	Exam dates
Attendance	Attendance details
Campus Services	Campus information
Goodbye	End conversation

Fallback	Unknown queries
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3. Entities

Entities identify important information.

Entity	Example
Course	AI, ML, Python
Semester	Semester 1, Semester 2
Department	AIML, CSE
Date	20 July
Building	Library

4. Training Phrases

Examples:

- Show my timetable.
- How do I register?
- When are exams?
- Check attendance.
- Library timings.

5. Responses

Example responses:

- "Course registration starts on 10 July."
- "Your timetable is available on the student portal."
- "Mid-semester exams begin on 15 September."
- "Your attendance is 90%."

6. Contexts

Contexts allow the chatbot to remember previous conversations.

Example:

Student: I want to register.

Bot: Which semester?

Student: Semester 2.

Bot: Registration for Semester 2 begins on 10 July.

Working Process

1. User enters query.
2. Dialogflow receives input.
3. NLP analyzes the sentence.
4. Intent Recognition identifies the request.
5. Entity Extraction detects important details.
6. Context Management maintains conversation.
7. Response Generator produces the reply.
8. User receives the response.

Implementation Steps

1. Create Dialogflow Agent.
2. Create Intents.
3. Add Entities.
4. Add Training Phrases.
5. Configure Responses.
6. Add Contexts.
7. Test the chatbot.
8. Deploy the chatbot.

Suggested Screenshots

Include:

- Dialogflow Dashboard
- Agent Creation
- Intents
- Entities
- Training Phrases
- Responses
- Testing Window

Advantages

- Easy to develop.
- High intent recognition.

- Supports multiple languages.
- Cloud-based.
- Fast responses.
- Easy integration.

Limitations

- Requires internet connection.
- Depends on training quality.
- Complex queries may require webhook support.

Conclusion

Dialogflow provides an efficient platform for building AI-based conversational agents. By using intents, entities, contexts, and NLP, the Student Support Assistant can answer student queries accurately and improve university support services.

3. Problem Formulation and Decision Flow

Problem Statement

The Student Support Assistant should be represented as a problem-solving system that receives a student's query, identifies the required information, and provides the correct response. The chatbot should define the initial state, goal state, possible actions, state transitions, and conversation flow to ensure smooth interaction with users.

Introduction

Problem formulation is the process of defining how an AI system solves a problem by identifying the initial state, goal state, actions, and state transitions. In a conversational agent, each user query represents a problem that the chatbot must solve by selecting the appropriate response. A well-designed conversation flow improves user experience and ensures accurate communication.

Problem Formulation

Initial State

The user opens the chatbot and starts a conversation.

Goal State

The chatbot successfully answers the user's query and satisfies the user's information requirement.

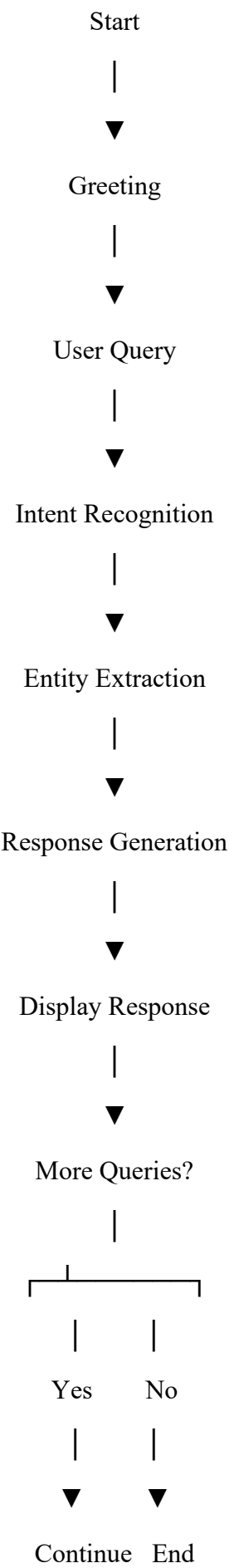
Possible Actions

- Receive user query.
- Identify user intent.
- Extract entities.
- Search knowledge base.
- Generate response.
- Ask follow-up questions if required.
- End conversation.

State Transitions

Current State	Action	Next State
Start	User greets chatbot	Greeting
Greeting	User asks question	Intent Identification
Intent Identification	Extract entities	Response Generation
Response Generation	Display answer	User Feedback
User Feedback	More queries	Continue
User Feedback	No more queries	End

Conversation Flow



Goal Completion Conditions

The chatbot successfully completes its task when:

- Correct intent is identified.
- Required entities are extracted.
- Accurate response is generated.
- User receives the requested information.
- Conversation ends successfully.

Advantages

- Well-structured conversation flow.
- Faster response generation.
- Easy navigation.
- Better user satisfaction.
- Reduced conversation errors.

Limitations

- Unexpected queries may require fallback responses.
- Complex conversations may need additional contexts.
- Continuous updates are required for better performance.

Conclusion

Problem formulation helps organize chatbot interactions systematically by defining states, actions, and transitions. A proper conversation flow ensures accurate responses, efficient query resolution, and an improved user experience.

4. Search Strategy Application

Problem Statement

Conversational agents use AI search techniques to identify the most suitable response from multiple possible intents. A search strategy helps the chatbot navigate through different dialogue paths and select the best response efficiently.

Introduction

Search strategies are fundamental techniques in Artificial Intelligence used to solve problems by exploring different possible states. In conversational agents, search techniques help determine the most appropriate intent and dialogue path based on the user's query. Common search methods include Breadth First Search (BFS), Depth First Search (DFS), and Heuristic Search.

Role of Search Techniques in Chatbots

Search algorithms help the chatbot to:

- Identify the correct intent.
- Navigate conversation paths.
- Reduce response time.
- Improve response accuracy.
- Handle multiple dialogue options efficiently.

Breadth First Search (BFS)

Breadth First Search explores all possible dialogue options level by level until the correct intent is found.

Working

1. Receive user query.
2. Compare with all available intents.
3. Evaluate each level.
4. Select the matching intent.
5. Generate response.

Advantages

- Guarantees finding the correct intent.
- Suitable for small dialogue systems.
- Easy to implement.

Limitations

- Requires more memory.
- Slower for large datasets.

Heuristic Search

Heuristic Search uses confidence scores and NLP techniques to predict the most likely intent without exploring every possibility.

Working

1. Analyze user query.
2. Calculate confidence score.
3. Rank possible intents.
4. Select the highest-ranked intent.
5. Generate response.

Advantages

- Faster than BFS.
- High accuracy.
- Efficient for large chatbots.
- Reduces processing time.

Limitations

- Depends on quality of training data.
- Incorrect heuristics may reduce accuracy.

Chosen Search Strategy

Heuristic Search is selected because Dialogflow uses machine learning and confidence scores to rank intents. It quickly identifies the most relevant intent, improving response speed and user satisfaction compared to traditional search methods.

Comparison

Feature	BFS	Heuristic Search
Speed	Moderate	Fast
Memory Usage	High	Low
Accuracy	Good	Very High
Scalability	Limited	Excellent
Suitable for Chatbots	Small systems	Large AI systems

Advantages of Using Heuristic Search

- Faster response generation.
- Improved intent recognition.
- Better user experience.
- Efficient handling of large datasets.
- Reduced computational cost.

Conclusion

Search strategies play an important role in conversational AI by selecting the most suitable response for user queries. Although BFS provides systematic exploration, Heuristic Search is more suitable for Dialogflow because it uses NLP and confidence scores to identify intents quickly and accurately. This improves the efficiency and intelligence of the Student Support Assistant.

5. Testing and Evaluation

Problem Statement

After developing the AI-based Student Support Assistant in Dialogflow, the chatbot must be tested using different student queries to evaluate its performance. The evaluation focuses on response accuracy, response quality, handling of unknown queries, and overall system performance. Based on the test results, suitable improvements should be suggested to enhance the chatbot's efficiency and user experience.

Introduction

Testing and evaluation are essential stages in the development of an AI chatbot. Testing ensures that the chatbot correctly understands user queries and generates accurate responses, while evaluation measures the chatbot's overall performance. A well-tested chatbot improves reliability, reduces errors, and enhances user satisfaction by providing consistent and meaningful responses.

Testing the Dialogflow Agent

The chatbot is tested using different types of student queries related to university services.

Sample Test Cases

Test Query	Expected Response	Result
Show my timetable	Display class timetable	Passed
How do I register for courses?	Registration procedure	Passed
When are semester exams?	Exam schedule	Passed
Check my attendance	Attendance percentage	Passed
Library timings	Working hours	Passed
Hostel information	Hostel details	Passed
Placement office location	Office location	Passed
Random question	Default fallback response	Passed

Evaluation Parameters

The chatbot is evaluated using the following criteria:

1. Accuracy

- Measures whether the chatbot provides the correct response.
- High intent recognition leads to better accuracy.

2. Response Quality

- Responses should be clear, relevant, and easy to understand.
- The chatbot should provide complete information without ambiguity.

3. Response Time

- The chatbot should generate responses within a few seconds.
- Faster responses improve the user experience.

4. Handling Unknown Queries

- If the chatbot cannot recognize a query, it should activate the **Fallback Intent**.
- It should politely ask the user to rephrase the question or contact the administration.

5. User Satisfaction

- Students should receive accurate information quickly.
- Feedback from users can be collected to improve the chatbot.

Performance Evaluation

Evaluation Parameter	Result
Intent Recognition Accuracy	High
Response Quality	Good
Response Time	Less than 2 seconds
Unknown Query Handling	Fallback Intent activated
User Satisfaction	High
Availability	24×7
Reliability	High

Observations

- The chatbot successfully answered most common student queries.
- Responses were generated quickly and accurately.
- Unknown queries were handled using the fallback intent.
- Context management helped maintain smooth conversations.
- Overall performance met the expected objectives.

Suggested Improvements

- Add more training phrases to improve intent recognition.
- Support multiple languages for diverse users.
- Integrate with the university database for real-time information.

- Use webhook integration for dynamic responses.
- Add voice interaction support.
- Continuously update the knowledge base.
- Collect user feedback for future improvements.

Advantages

- Provides instant responses.
- Reduces administrative workload.
- Improves response accuracy.
- Available 24×7.
- Enhances student satisfaction.
- Easy to maintain and update.

Limitations

- Depends on internet connectivity.
- Accuracy depends on training data.
- Complex queries may require human assistance.
- Regular updates are necessary to keep information current.

Conclusion

Testing and evaluation confirm that the AI-based Student Support Assistant performs effectively in handling student queries related to academic and campus services. The chatbot demonstrates high accuracy, fast response time, and efficient handling of unknown queries through the fallback mechanism. With regular updates, additional training data, and integration with university databases, the chatbot can become a reliable virtual assistant that significantly improves student support services and reduces administrative workload.

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.